

sensor input to the circuit of Fig. 1. This makes the power available to the base of transistor T1, which now conducts. This makes transistors T2 and T3 conduct. Thus the power supply is available for the melody circuit (Fig. 1).

The melody generator IC is programmed to produce twelve different tunes repeatedly until the shower water is running and transistors T1, T2

and T3 are conducting. The volume of melody generator tunes can be controlled through preset VR1.

When the shower is turned off, both transistors T1 and T2 cut off and a high voltage develops at the base of npn transistor T3. Thus transistor T3 stops conducting. This, in turn, cuts off the power supply to the melody generator circuit, which stops gener-

ating musical tunes. Since the circuit consumes virtually no power when the shower is turned off, the battery lasts long.

Assemble the circuit on a general-purpose PCB and enclose it in a plastic case with LED1 and switch S1 accessible from outside. Take the sensor wires and connect to the wires on shower at A and B ends. ●

## NIGHT LAMPS

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Here are two night lamp circuits using LEDs. One could be used as a night-vision clock and the other as a TV lamp. Both the circuits are AC operated

out the characteristic colour of white LEDs and the 15V zener diode maintains the voltage at a safe level.

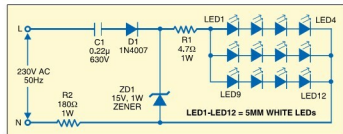


Fig. 1: Night-vision clock circuit

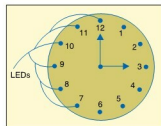


Fig. 2: LEDs mounted through drilled holes

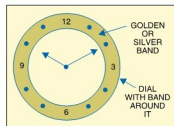


Fig. 3: LEDs mounted below the band

and consume very little power. These are also protected against mains fluctuations. The night-vision lamp uses twelve LEDs arranged in the circular pattern of a wall clock, while the TV lamp uses 24 LEDs in prism format.

Fig. 1 shows the circuit of night-vision clock. Capacitor C1 (0.22µF) reduces the current to safe limit, diode D1 provides rectified DC to bring

Resistor R2 protects the circuit against high inrush current, while resistor R1 is used for additional protection of the LEDs.

Assemble the circuit on a general-purpose PCB and mount the white LEDs (LED1 through LED12), through drilled holes, in the circular format of the clock dial as shown in Fig. 2. Similarly, you can mount LEDs on another

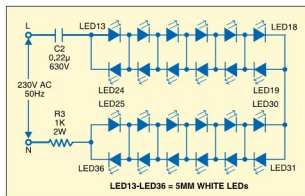


Fig. 4: TV lamp circuit

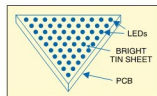


Fig. 5: LEDs mounted in a triangular format on a general-purpose PCB

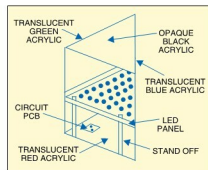


Fig. 6: TV lamp fittings

dial of the clock as shown in Fig. 3.

Fig. 4 shows the TV lamp circuit. Watching TV in darkness puts strain on your eyes. So an illumination of 10-15W is recommended. The circuit described here is known as Ken Schultz circuit. It uses a total of 24 white LEDs (LED13 to LED36) and is well balanced against surges and voltage spikes. The lamp produces almost no heat.

Assemble the circuit on a general-purpose PCB, with the white LEDs (LED13 through LED36) mounted in a triangular format (Fig. 5). Now enclose it in the proposed TV lamp fitting shown in Fig. 6. ●